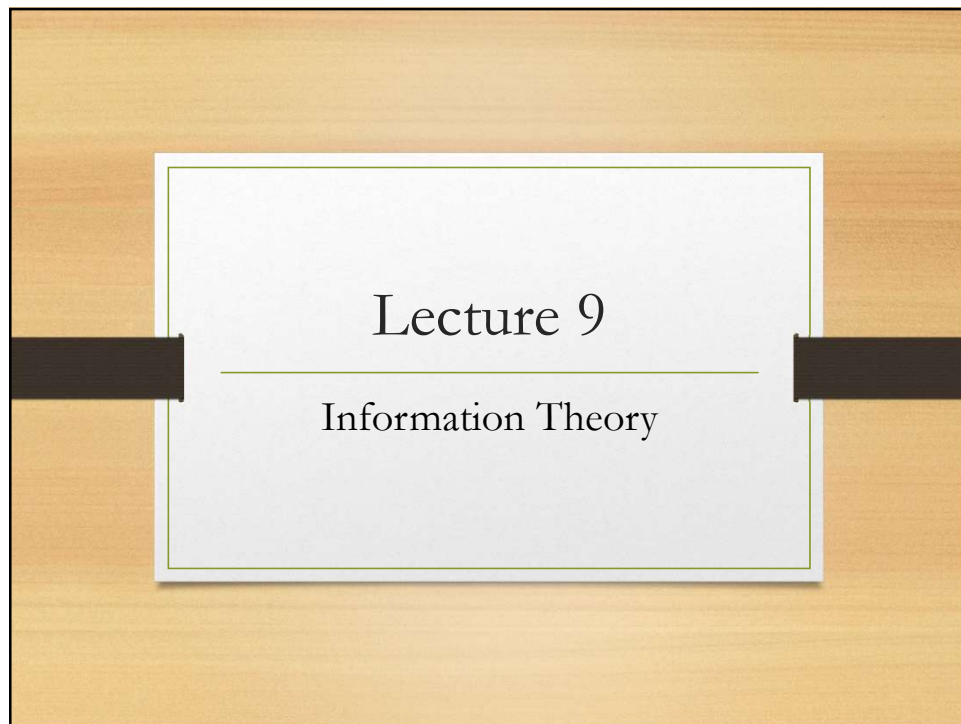
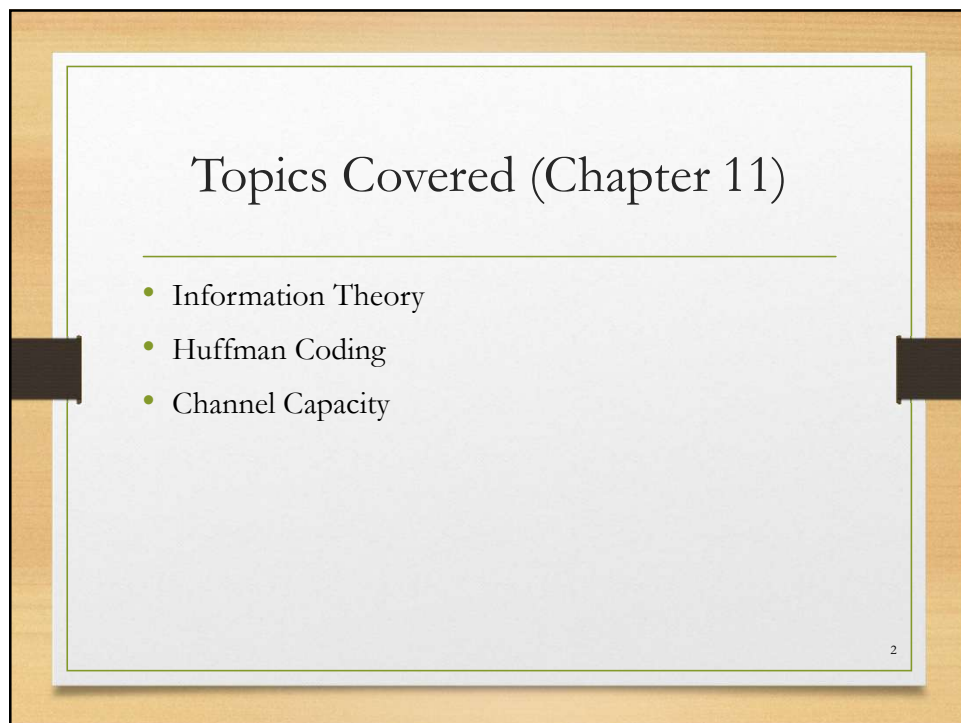




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Introduction

The purpose of a communication channel is –

- The transmission of information from point to point
 - High efficiency
 - High reliability
- Information theory provides a measure of the information in a message.
- The capacity of the channel can be determined.
- Efficient codes can be found.
- Errors can be managed with no loss of message.

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Information

- Consider the message –

Hoppy birthday Jobx.

This received message has errors, but since plain text has redundancy (and therefore is not efficient) it is possible to work out what the message should be

Happy Birthday John!

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Information

- This message –

§374621c

Contains no redundancy and we don't know if it is correct or incorrect.

- Information theory will help us choose a efficient, reliable code. Systematic redundancy can be added to the transmitted signal so that the channel can be used with improved reliability.

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Information Source

- A practical source can produce either analogue (continuous) or digital (discrete) messages.
- A discrete source will be considered as it is easier to quantify.
- A discrete source has a finite set of symbols (or letters) – the source alphabet.
- A discrete memoryless source (DMS) is one for which each symbol produced is independent of the previous symbol.

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Information Content

- The amount of information contained in an event is closely related to its uncertainty.
- Messages containing knowledge of high probability of occurrence convey little information.
 - *The sun will rise in the east.*
 - Does this message need all its symbols to convey this information ?
- Information should be proportional to the uncertainty of an outcome.

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Information Content

- Mathematically –

Consider now a discrete memoryless source (DMS), denoted by X , producing a finite number of message symbols $(x_1, x_2, x_3, \dots, x_m)$

The *information content* of symbol x_i , denoted by $I(x_i)$, is defined by

$$I(x_i) = \log_b \left(\frac{1}{p(x_i)} \right) = -\log_b (p(x_i)) \text{ bits/symbol}$$

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Self-Information

An information source is characterized by the following Alphabet and Probability Distribution:

Alphabet $A = \{a_1, a_2, a_3, a_4, a_5\}$

Probability Distribution $P = \{0.1, 0.3, 0.4, 0.15, 0.05\}$

Compute the **self-information** (in Bits) for all five source symbols.

$$I(a_1) = -3.32 \log_{10}(P(a_1)) = -3.32 \log_{10}(0.1) = 3.32 \text{ bits/ symbol}$$

$$I(a_2) = -3.32 \log_{10}(P(a_2)) = -3.32 \log_{10}(0.3) = 1.74 \text{ bits/ symbol}$$

- and so on

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Entropy

- In the case of a very large number of n symbols, the i^{th} symbol will occur np_i times and so the information received from the source is the summation of all such quantities.

$$\text{Total } I = \sum_{i=1}^m P(x_i) I(x_i)$$

- The mean value of $I(x_i)$ over the alphabet is known as the *entropy*

$$H(X) = \sum_{i=1}^m P(x_i) I(x_i)$$

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Example

- *An information source is characterized by the following alphabet and probability distribution -*

Alphabet $A = \{a1, a2, a3, a4, a5\}$

Probability Distribution $P = \{0.1, 0.4, 0.3, 0.15, 0.05\}$

Compute the Entropy for the source.

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Code Efficiency

- Code length – Let X be a DMS with finite entropy $H(X)$ and an alphabet $\{x_1, \dots, x_m\}$ with corresponding probabilities of occurrence $P(x_i)$ ($i=1 \dots m$). Let the binary code word assigned to symbol x_i by the encoder have length n_i measured in bits.

The average code word length –

$$L = \sum_{i=1}^m P(x_i) n_i$$

- Efficiency = $\eta = H(X)/L$

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Type of Code

- Fixed length code
 - Baudot code – length $L = 5$ bits per character, 32 character alphabet, used with the telegraph system.
 - ASCII code – length $L = 8$ bit code, 256 character alphabet, used with PCs.
 - reasonable simple to generate and simple to read with a machine.
 - Easy to detect errors.

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Binary codes

x_i	Code 1	Code 2	Code 3	Code 4	Code 5	Code 6
x_1	00	00	0	0	0	1
x_2	01	01	1	10	01	01
x_3	00	10	00	110	011	001
x_4	11	11	11	111	0111	0001

- Fixed length code
 - codes 1 & 2
- Variable length code
 - all codes except code 1 & 2.
- Prefix-free code
 - No code word can be formed by preceding a code symbols to another code word. Code2, 4 and 6.
- Optimal code
 - Has a minimum average length
- Uniquely decodable code.
 - Code 5 is but code 3 is not.

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Variable length code

- Variable length code is one whose code word length is not fixed.
- Prefix-free code is a code in which no code word can be formed by adding code symbols to another code word.
 - 0, 1, 00, 11 – is not a prefix-free code.
- A distinct code is uniquely decodable if the original source sequence can be reconstructed perfectly from the encoded binary sequence.
 - 0, 01, 011, 0111 – is a uniquely decodable code.

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Source Coding

- Source coding refers to the coding process that attempts to compress the data from a source by removing as much redundancy as possible. Sometimes it is also known as data compression.
- The objective of source coding is to use codes that are as short as possible to represent the source signal.
- Shorter codes are more efficient because they require less time to transmit at a given data rate.

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Source Coding

- One way to obtain efficient source coding is by assigning more likely messages with shorter source codes and less likely messages with longer source codes.
- Example: Morse code adopts various combination of dashes and dots (i.e., code words) are assigned to each letter. To minimize transmission time, shorter code words are assigned to more frequently occurring (more probable) letters such as e, t, and a; and longer code words are assigned to rarely occurring letters (less probable) such as x, q, and z. This explains why Morse code is a good source code.

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Entropy Coding

- For digital signals, the overall transmission time is minimized if a message (or symbol) of probability P is assigned a code word with a length proportional to $\log(1/P)$. This is known as entropy coding.
- Entropy coding techniques:
 - Huffman Coding
 - Arithmetic Coding
 - Lempel-Ziv Coding

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Huffman encoding

- The Huffman code is the optimal prefix and variable-length code given the symbol's probabilities of occurrence.
- The procedure for generating a Huffman code is –
 - List the source symbols in order of decreasing probability.
 - Combine the probabilities of the two symbols having the lowest probabilities and reorder the resultant probabilities. Repeat until there are two remaining probabilities.
 - Start encoding with the last reduction. Assign a 0 to the first symbol and a 1 to the second symbol.
 - Now go back and assign 0 & 1 to the second digit that were combined. Repeat.

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Huffman Coding

- A communication system uses the following source symbols –
 - X_1 and $P(X_1) = 0.30$,
 - X_2 and $P(X_2) = 0.05$,
 - X_3 and $P(X_3) = 0.12$,
 - X_4 and $P(X_4) = 0.20$,
 - X_5 and $P(X_5) = 0.08$,
 - X_6 and $P(X_6) = 0.25$
- Generate an optimum code.
- The Huffman code is optimal and therefore efficient.

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Procedures

1. List the symbols in order of decreasing probability.
2. Combine the probabilities of the two symbols with the lowest probability. Re-order the list.
3. This process is repeated until there are two ordered probabilities remaining.

X_i	$P(X_i)$
X_1	0.30
X_6	0.25
X_4	0.20
X_3	0.12
X_5	0.08
X_2	0.05

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X_i	$P(X_i)$	Reduction 1	Reduction 2	Reduction 3	Reduction 4
X_1	0.30	0.30	0.30	0.45	0.55
X_6	0.25	0.25	0.25	0.30	0.45
X_4	0.20	0.20	0.25	0.25	
X_3	0.12	0.13	0.20		
X_5	0.08	0.12			
X_2	0.05				

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Procedures (cont'd)

4. Start encoding with the last reduction (right-hand side of table).
5. Assign 0 as the first digit in the code word for all symbols associated with the first probability. Assign 1 to the second probability.
6. Work your way back to the left-hand side of the table adding in binary digits as required.
7. Keep working back until the first column is reached.

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X_i	$P(X_i)$	Reduction 1	Reduction 2	Reduction 3	Reduction 4
X_1	0.30 00	0.30 00	0.30 00	0.45 1	0.55 0
X_6	0.25 01	0.25 01	0.25 01	0.30 00	0.45 1
X_4	0.20 11	0.20 11	0.25 10	0.25 01	
X_3	0.12 101	0.13 100	0.20 11		
X_5	0.08 1000	0.12 101			
X_2	0.05 1001				

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Lempel-Ziv Coding

- Until now, symbol distributions were known in advance.
- What happened if they are not known?
 - Lempel-Ziv coding solves this general case, where only a stream of bits is given.
 - LZ coding creates its own dictionary (string of bits), and replaces future occurrences of these strings by a short position string.

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LZ coding

- Parses source input (in binary) into the shortest distinct strings:

1011010100010 => 1, 0, 11, 01, 010, 00, 10
- Each string includes a prefix and an extra bit (010 = 01 + 0), therefore encoded as: (prefix string place, extra bit)
- Requires 2 passes over the input (one to parse input, second to encode). Can be modified to one pass.

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LZ coding Steps

- 1. Initialize the dictionary to contain an empty string ($D=\{\phi\}$).
- 2. $W \leq$ longest block in the input string which appears in D .
- 3. $B \leq$ first symbol in the input string after W
- 4. Encode W by its index in the dictionary, followed by B
- 5. Add $W+B$ to the dictionary.
- 6. Go to Step 2.

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LZ Coding—an Example

1011010100010...=>1, 0, 11, 01, 010, 00, 10, ...

Index	D	0	1	2	3	4	5	6	7
0	ϕ	String	1	0	11	01	010	00	10
1	1	W	ϕ	ϕ	1	0	01	0	1
2	0	B	1	0	1	1	0	0	0
3	11	pair	(0,1)	(0,0)	(1,1)	(2,1)	(4,0)	(2,0)	(1,0)
4	01	coding	0001	0000	0011	0101	1000	0100	0010
5	010								
6	00								
7	10								
8	...								



0001000000110101100001000010...

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Channel Capacity (Noiseless Channel)

- A maximum of $2B$ independent pieces of information per second can be transmitted, error free, over a noiseless channel of bandwidth B Hz. This follows from the sampling theorem.

$$C = 2B \log_2 M$$

- The sampling theorem shows that a low-pass signal of bandwidth B Hz can be fully recovered from samples uniformly taken at the rate of $2B$ samples per second.

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Channel Capacity (Noisy Channel)

- In 1948 Claude Shannon published *A Mathematical Theory of Communication*.
- This work focuses on the problem of how best to encode the information a sender wants to transmit. In this fundamental work he used tools in probability theory.
- Shannon developed information entropy as a measure for the uncertainty in a message while essentially inventing the field of information theory.

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Shannon – Hartley Theorem

- The Shannon–Hartley theorem states that the channel capacity C , meaning the theoretical maximum rate of clean data that can be sent with a given average signal power S through an analogue communication channel subject to noise of power N , is:

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

where

C is the channel capacity in bits per second;

B is the bandwidth of the channel in hertz;

S is the total signal power over the bandwidth, measured in watt or volt²;

N is the total noise power over the bandwidth, measured in watt or volt²;

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